

Concept Title:

ROMEO - Regolith Oxygen Magnetic Extraction Operative

Inspiration for your Concept:

We aim to introduce an unprecedented approach to collecting and transporting lunar regolith for oxygen extraction, leveraging the magnetic properties of regolith. Our innovative solution not only facilitates oxygen extraction but also enhances overall support for sustaining human presence on the Moon and beyond by extracting other useful elements.

Executive Summary:

Create a brief introduction to your concept.

ROMEO, which stands for Regolith Oxygen Magnetic Extraction Operative, is a groundbreaking lunar excavation solution set to revolutionise lunar regolith collection for oxygen extraction. By leveraging the magnetic properties of lunar regolith, which contains oxygen-rich, ferrous materials, ROMEO's magnetic extraction provides an additional filter for the regolith, ensuring unparalleled efficiency in later oxygen extraction.

In essence, ROMEO is a brushless motor-driven circular drum with six strategically positioned ferromagnetic solenoids, ensuring precise regolith attraction and collection. Linear rails, combined with a LIDAR sensor, provide consistent lowering of the system and incorporate feedback loops to maintain a reliable 1-5mm distance above the lunar surface during excavation. After just 5 revolutions, ROMEO's hopper will conservatively hold 219g of regolith, and a linear actuator trap door system can then be employed to deposit the regolith in a controlled manner.

This excavation mechanism seamlessly integrates with the Trailblazer rover, weighing a mere 1.75kg and occupying only approximately 11% of the available volume. Overall, ROMEO's unique design ensures lunar stability, marking a paradigm shift in regolith acquisition technology.

As ROMEO progresses through the active design phase at Technology Readiness Level 3, it emerges as a creative and efficient solution, blending electromagnetism and cutting-edge engineering to address the lunar regolith collection challenge with unparalleled ingenuity.

Concept Description:

Provide a description of your concept and how it will enable a rover to collect regolith on the lunar surface. Describe the concept's main components, and how it would integrate with a lander (e.g., whether the solution is a standalone attachment, part of a rover or a combination of both, and whether it requires any inputs such as data, movement or power). Finally, discuss the strengths and weaknesses of your approach, including any technical risks.

NASA research has found that the regolith on the moon is composed of 40-45% oxygen by weight but requires extreme temperatures for extraction. This is due to oxygen molecules being bound within compounds (such as silicates and oxides) with robust chemical bonds. The primary oxides in lunar regolith are ilmenite (FeTiO_3) and Iron (II) Oxide (FeO), both

ferromagnetic materials. Therefore, our solution, ROMEO, ingeniously utilises these magnetic characteristics to collect regolith.

ROMEO is an extraction mechanism mounted on a plate connected to linear rails externally affixed to the lunar rover's rear. It consists of a brushless motor-driven circular drum with six equally spaced solenoids. Adjacent to ROMEO is a hopper. As the motor rotates the drum, powered solenoids, acting as electromagnets through a slip ring, attract the magnetic regolith oxygen compounds. The combination of a hall effect sensor and motor encoder allow us to know the different solenoid locations, meaning we can deactivate them at certain positions for deposition into the hopper and reactivate them for collection. This process repeats until the 219g hopper is full, conservatively estimated to be after approximately 5 full rotations. Linear rails facilitate lowering ROMEO to approximately 1 - 5 mm above the lunar surface. ROMEO operates optimally by driving at slow speeds, utilising a LIDAR sensor to maintain a consistent 1-5mm height off the ground. After collection, linear rails raise ROMEO, providing sufficient clearance for the rover to manoeuvre uneven terrain. During deposition, the hopper lowers to an appropriate height above NASA's facility run ISRU deposition box. Utilising a linear actuator-driven linkage, a trap door beneath the hopper opens for a controlled regolith deposition.

To ensure the attractiveness of ferrous materials, the solenoid core is made of 99.5% annealed iron. This material selection massively increases the effectiveness of the solenoid compared to a vacuum core, with annealed iron being 200,000x more powerful due to the relative permeability of the core material.

What makes this a good design is its ability to collect oxygen-containing materials, the mission's primary objective, with the additional benefit of gathering useful metals that could also be used to enhance our lunar presence. Using a magnetic excavation method compliments molten regolith electrolysis, resulting in far more efficient oxygen extraction. This efficiency is due to the fact that a magnetic extraction would minimise silicate collection which is more power-intensive during electrolysis for oxygen extraction compared to oxides. However, this design requires more on-board power consumption than conventional mechanisms like a scraper, though deposition is likely less power-intensive than a typical bucket drum. The linear rail-mounted ROMEO near the rover's centre of mass enhances stability without significantly altering the moment, unlike designs with rotating arms. This is advantageous in low gravity, preventing tipping by employing shallow magnetic "scrapes" rather than forceful movements. However, the design complexity demands technical expertise in development and experimentation, and more space-proofing is essential due to multiple moving parts which can add to its weight.

Novelty / Creativity:

Describe how your system demonstrates novelty or a new application of existing concepts within the constraints of the rover and lunar environment.

ROMEO represents a significant departure from traditional excavation mechanisms by utilising electromagnetism for regolith collection, positioning itself as a novel and creative solution tailored to the distinctive constraints of rovers and lunar environments. In contrast to

commonplace mechanical systems, often reliant on scraping or digging, ROMEO's method minimises direct and forceful contact with the lunar surface. The distinctive use of electromagnets enables ROMEO to selectively attract and capture regolith specifically containing oxygen compounds, illustrating a new application of existing concepts designed and tailored for the unique constraints of the lunar environment. Unlike contemporary excavation mechanisms that lack a personalised touch to the lunar surface, relying on physical excavation tools, ROMEO's innovative approach not only enhances operational efficiency—by facilitating the collection of more useful compounds that require less energy to breakdown—but also mitigates the potential for wear and tear commonly associated with mechanical components digging into the abrasive and destructive ground. Thus, ROMEO stands out as a creative adaptation of existing principles, showcasing the integration of electromagnetism as a more efficient and sustainable means of regolith collection within the challenging lunar environment.

Functionality:

Explain how your system is designed to fit with a rover and its operating environment on the lunar surface, given the constraints outlined in the challenge description. Include details about assumptions, mass, size considerations, and how it meets the mission's objectives.

In alignment with the lunar rover's constraints, ROMEO is designed to seamlessly integrate onto the rover's rear, mounted on linear rails. Alternatively, it can be positioned at the front or underneath the rover, contingent on ground clearance, rover design and potential adjustments to the actuating system. Weighing a mere 1.75kg, less than 10% of the rover's total weight, ROMEO adheres to the 20kg restriction imposed on the Trailblazer rover. Ideally, ROMEO would be externally mounted close to the rover's cross-bar and centre of mass (COM) for added stability, however, due to its lightweight, it should still maintain stability, even when mounted away from the COM.

Additionally, the assembly's 188.4mm x 172.5mm x 272.7mm stowed dimensions easily fit within NASA's 500mm x 400mm x 400mm requirements, occupying only ~11% of the available volume.

Moreover, the powered linear rails secure ROMEO and facilitate lowering the mechanism to a specified distance from the lunar surface, ranging from 1-5mm. This flexibility ensures adaptive passes over the lunar terrain without compromising mission objectives. Furthermore, the linear rails allow ROMEO to be completely raised, making it suitable for extended travel without issues. Additionally, the lowering capability enables controlled deposition into the NASA ISRU facility.

Assuming a 1kg regolith collection over 5 days, the hopper was designed to conservatively hold 219g of regolith. This size signifies that only one trip is required per day - with a 1.1 safety factor. These conservative estimates suggest that the system designed can potentially hold more.

The moderate power consumption strikes a strategic balance, ensuring ROMEO's continuous functionality within acceptable levels. With only approximately five revolutions needed to fill

the hopper, the system efficiently utilises battery power, aligning with the rover's capability to supply short bursts of energy during the collection process. Assumptions about sufficient current for the solenoids are grounded in extensive empirical testing required, considering the unique lunar environment. The low lunar gravity, coupled with the high magnetic field power from the iron solenoid inner material, enhances the system's ability to lift regolith effectively. Additionally, the systematic and simple operation of the solenoids, featuring a repeatable on-off state, ensures consistent and reliable functionality.

Durability:

Describe how your system ensures durability, including its capacity for remaining intact after multiple acquisition and deposit cycles, without requiring regular servicing or maintenance. Explain how it overcomes challenges on the lunar surface and maintains consistent performance.

Due to the abrasive nature of lunar regolith, durability was a critical consideration in finalising ROMEO as an excavation system. Unlike other systems relying on cutting blades or teeth for regolith extraction, ROMEO's utilisation of electromagnetism minimises forceful interactions with the lunar surface. This design choice significantly reduces the impact of regolith abrasion on the external part of the drum, ensuring the longevity and consistency of ROMEO throughout the collection process. As a result, this approach guarantees consistent and reliable acquisition performance, meeting the criterion for durability.

Addressing the critical concern of preventing regolith from entering the system, our design incorporates essential features such as O-rings and a protective cover over any moving parts. These measures are pivotal in maintaining the integrity of ROMEO over multiple acquisition and deposit cycles. Furthermore, the inclusion of a LIDAR sensor ensures a consistent height during operation, contributing to the system's consistency and reliability in acquiring and depositing regolith over numerous cycles.

Additionally, ROMEO is mainly made of 7075 aluminium, renowned for its exceptional strength-to-weight ratio and corrosion resistance, ensuring the system withstands the harsh lunar conditions while remaining lightweight. The strategic choice of robust yet efficient brushless motors in ROMEO's design also contributes to its longevity, minimising wear and tear over multiple lunar regolith acquisition and deposit cycles.

Despite the slightly increased power requirement for the electromagnetism process compared to other mechanical excavation methods, we assert that the Australian rover's sole focus on regolith collection signifies that it is still very appropriate and does not aimlessly expend any finite resources. Instead, it ingeniously utilises the inherent magnetic properties of lunar regolith compounds, optimising its functionality and allowing for less power required when performing oxygen extraction. Overall, ROMEO's unique design means regular servicing or maintenance is not required due to its efficiency in overcoming the challenges posed by the lunar environment.

Scalability:

Discuss any potential constraints that may prevent your system from being scaled up. Identify and address any manufacturing, material, or mass issues that may limit scalability.

Scaling up ROMEO may introduce manufacturing challenges, particularly with larger components such as solenoids or custom machined parts. While the manufacturing process for such elements may need adjustment, we view it as a manageable aspect. For example, the world's largest fusion reactor in France utilises a central solenoid measuring around 2.1m tall and 4.3m wide, demonstrating that manufacturing large-scale solenoids is feasible.

Custom machined parts, typically produced on a lathe, may require additive manufacturing when scaled up - depending on the scale. Despite this variation, we remain confident in overcoming manufacturing challenges, with the process potentially changing (depending on the scaling factor) but the end result remaining the same.

Additional potential limitations on scalability may involve materials. However, our intention to use materials such as aluminium and copper wire, known for versatility in various-sized applications, alleviates concerns about scalability limitations.

Concerning mass, while scaling up may increase weight, the relatively low density and high strength of 7075 aluminium—our most commonly used material—compared to other materials ensure that when scaling the entire rover, it remains a manageable factor.

A notable consideration is the potential non-linear relationship between the system's scaled size and the power required. Despite this, we maintain confidence, anticipating a manageable adjustment in power requirements during scaling.

Evidence of Design Optimisation:

Provide evidence of critical evaluation during the design process. Explain how design decisions were justified and how you optimised the design from concept to the final solution.

Throughout the design process, we meticulously evaluated the problem by methodically breaking down the brief, defining objectives, functions, and specifications. Our critical evaluation began with extensive research on Lunar and Martian missions, diverse excavation methods, material choices in space, and mounting possibilities - drawing insights from academic research. To ensure well-informed decision-making, we developed a comprehensive criteria framework with weighted factors. Generating and evaluating six concepts against established criteria, the magnetic excavator emerged as the statistically significant choice, inspiring the subsequent design phase.

During ROMEO's CAD development, we optimised design elements. Utilising brushless motors as inspiration, we strategically chose six solenoids for practical wiring. Incorporating a slip ring prevented cable entanglement, while meticulous pocketing reduced unnecessary weight. Design refinements, including ceramic bearings and spacers for motor stability, O-rings, rubber seals, and welded-shut enclosures for dust-proofing, along with an intricately designed hopper integration made of lightweight sheet metal.

Orientation decisions and lowering methods were based on proximity to the rover's centre of mass, stability, and machinability. To address past space mission challenges, we integrated carefully positioned cameras, enhancing overall functionality.

Material selection involved a rigorous process, adhering to criteria such as a low thermal coefficient, high strength-to-weight ratio, machinability, and ease of sourcing, like high-tensile and corrosion-resistant 7075 aluminium.

Technology Readiness:

Discuss your assessment of the technical maturity of your concept. Assign a Technology Readiness Level (TRL) to it, and provide a reasoned explanation or supporting evidence for the chosen rating. Note: a description of TRL can be found [here](#)

Due to the novelty of ROMEO in excavation collection, ROMEO is positioned at Technology Readiness Level (TRL) 3. Electromagnetism, created in the 19th century, has continuously evolved, and the study of regolith from the Apollo missions revealed its oxygen and iron content. This accumulation of knowledge has satisfied TRL 1. Electromagnetism has undergone ongoing refinement and widespread application, from everyday devices like brushless motors to advanced technologies like maglev. Regolith has also been further investigated, with its electrically charged and magnetic nature being discovered. Our solution ingeniously combines two well-developed concepts into a new, unique process and environmental conditions, satisfying TRL 2. Currently at TRL 3, ROMEO's active design phase has commenced, with research planning in progress. Although no specific experimental proof-of-concept has yet been established, the design process is underway, and further validation and testing are crucial to progress through higher TRLs, acknowledging the preliminary stage of ROMEO's development.